MIL-F-2098D (MU)
Amendment 3
30 November 1971

Superceding
Amendment 2
20 September 1971

MILITARY SPECIFICATION

FUZE, PD, M503A2, LESS PELLETS
BOOSTER AND BOOSTER LEAD, LOADING
ASSEMBLING, AND PACKING AND METAL PARTS FOR

This Amendment forms a part of Military Specification MIL-F-2098D, 20 September 1968, and is mandatory for use by Department of Army.

Page 3 - Para 3.9.1 - Line 3 - Delete 100 and substitute "400".

Page 3 - Para 3.9.2 - Delete last sentence and substitute "The angle of impact shall be approximately 5 degrees (°)

Page 3 - Para 3.9.3 - Line 3 - Deleted loaded 60 feet and substitute "located 10 feet"...

Page 3 - Add Paragraph 3.12

3.12 Transportation Vibration.-The fuze shall comply with the requirements of MIL-STD-331. The samples used shall be those selected for ballistic testing.

Page 4 - Paragraph 4.2.1 - Line 4 - Delete "50 assemblies" and substitute "200 assemblies".

Page 5 - Paragraph 4.2.1 - Delete lines 2 through 4, "...every subassembly...which have been"

Substitute: "...every subassembly, and noting that parts per drawing 9215034, 9215047, and 8797787 are loaded explosive components, and one hundred fifty (150) assemblies per drawing 9234644 which have been subjected to transportation vibration shall be furnished to the proving ground for ballistic testing, which have been..."

Page 5 - Paragraph 4.2.2 - Line 4 - Delete "4.4.10" and substitute "4.4.12".

FSC-1390

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Page 13 - Paragraph 4.3.2.13 - Delete "AQL 2.5 percent" and substitute "AQL 4.0 percent"

Page 13 - Add Minor 208:

208.	Raised metal present in scribe line	Visual	13021

Page 15 - 4.3.2.16 Add Major 104:

104.	Absence of green phosphorescent paint in the scribe line	Visual	16008

Page 15 - Delete Para 4.3.2.17 and substitute Para 4.3.2.17 as follows:

4.3.2.17 Rotor housing assembly (see dwg. 9215047 covering a detail of dwg. 9234644) prior to assembly of set back spring, setback sleeve, halfblock retainers and detent spring around the halfblock retainers.

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Four detent pins do not engage rotor	Visual-Manual	17001
2.	Detent spring restraining detent pins missing	Visual	17008
3.	Detent spring not locked in detent groove	Visual-Manual	17010
Major:	None Defined		
Minor:	None Defined		

Page 15 - Add Para 4.3.2.17.1

4.3.2.17.1 Rotor housing assembly (see dwg. 9215034 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Detent spring retaining halfblock retainers missing	Visual	17002
2.	Halfblock retainer(s) inverted	Visual	17003
Major:	AQL 0.40 percent		
101.	Threads damaged	Visual	17004
102.	Absence of green phosphorescent paint in scribe line	Visual	17006

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Minor:
201. Evidence of poor workmanship (see 3.11)...Visual 17005

Page 19 - 4.3.2.25 Add:

Major: AQL 0.40 percent

101. Fuze not sealed in polyethylene bag .....Visual 25007

Page 19 - 4.3.2.25 Add Minor:
203. Absence of stake, rotor housing assembly to body assembly .....Visual 25015

Page 19 - "4.3.3 Testing" add after "(see table II for disposition of samples after test.)"

Page 19 - 4.3.3.2 Delete and substitute:

4.3.3.2 Non-arming of rotor housing assembly (see dwg. 9215034) Critical defect - Code No. 17011. This test shall be performed 100 percent, on every lot. Any assembly which fails to comply with the requirements specified in paragraph 3.7 shall be classed defective and removed from the lot. As an additional inspection requirement the four detents when viewed via Fluoroscope or X-ray shall positively engage the rotor ball and all detents shall be within a template circle whose maximum diameter does not exceed 0.859 inches. Each housing assembly shall be marked as it exits from the equipment to verify that it has been tested. The accuracy of the inspection shall be verified hourly by passing standards through the equipment. One standard shall contain a missing detent, and the other shall contain a detent which protrudes beyond the maximum diameter of the detent circle of 0.859 inches. Failure to properly disposition the standards shall result in rejection of all material previously accepted since the last standard was properly identified. The test shall be the last test performed on the rotor housing assembly prior to insertion in the body and shall be performed as specified in 4.4.2 using test equipment in accordance with 4.3.4.

Page 19 - 4.3.3.3 Add after housing assembly "see dwg. 9215034."

Page 20 - Para 4.3.3.10 Add after last sentence. See Table I for Ballistic Test Requirements Summary.

Page 21 - Para 4.3.3.10.1 Delete and substitute:

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4.3.3.10.1 Non-Arm Check Test.-Forty (40) samples shall be selected from each lot for this test. If one or more samples function upon impact or one or more samples prematurely function, the lot shall be rejected. This test shall be performed as specified in 4.4.9.1. After three consecutive production lots have complied with 3.9.3, twenty (20) samples shall be selected from each lot for this test. If one or more samples function upon impact, or one or more samples prematurely function, the lot shall be rejected. This test shall be performed as specified in 4.4.9.1.

Page 21 - Para 4.3.3.10.2 Delete and substitute:

4.3.3.10.2 Ballistic Functioning.-Eighty (80) samples shall be selected from each lot for this test. If three (3) or more samples fail to function, or one or more samples prematurely function, the lot shall be rejected. This test shall be performed as specified in 4.4.9.2. After three (3) consecutive production lots have complied with 3.9.1, thirty (30) samples shall be selected from each lot for this test. If two (2) or more samples fail to function, or one or more samples prematurely function, the lot shall be rejected. This test shall be performed as specified in 4.4.9.2.

Page 21 Para 4.3.3.10.3 Delete and substitute:

4.3.3.10.3 Ballistic graze functioning.-Thirty (30) samples shall be selected from each lot for this test. If three (3) or more samples fail to function, or one or more samples prematurely function, the lot shall be rejected. The test shall be performed as specified in 4.4.9.3.

Page 21, Add Table I following paragraph 4.3.3.10.3.

TABLE I
BALLISTIC TEST REQUIREMENTS SUMMARY

Premature Burst (See Para 3.9).-Critical Defect. Any premature burst shall be cause for rejection of the lot.

Functioning (see para 3.9).-Major Defect

WEAPON	TARGET	TARGET DISTANCE	PREPRODUCTION FIRST THREE CONSECUTIVE SUCCESFUL LOTS SAMPLE SIZE	PRODUCTION TESTING SAMPLE SIZE
57mm,CTG M306A1	1 in. Plywood	10 ft.	40	20

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HE/Spotting Charge 57mm,CTG M306A1	1 in. Plywood	400 Ft.	80	30
HE/Spotting Charge 57mm,CTG M306A1 HE	Firm Ground	1,000 Ft*	30	30

*Angle of impact shall not exceed five (5) degrees

ACCEPTANCE CRITERIA
SUMMARY

(a) Preproduction And First Three Consecutive Successful Lots

	Accept	Reject	Code No.
Premature Burst	0	1	30001
Screen Duds	2	3	30002
Graze Duds	2	3	30003

(b) Production Testing After Three (3) Consecutive Successful Lots

	Accept	Reject	Code No.
Premature Burst	0	1	30004
Screen Duds	1	2	30005
Graze Duds	2	3	30006

Page 21 - Add paragraph 4.3.3.12:

4.3.3.12 Static Non Arm Test (see dwg. 9234644;-the fuze shall be tested less pellets, cup booster and cup booster lead. Critical defect, Code No. 25008. This test shall be performed one hundred (100) percent. The fuze shall not be spun when subjected to this test. Any one (1) assembly which fails to pass this test shall be classed defective, and shall be removed from the lot. These rejects may be reworked, but cannot be replaced in the original lot. They can be formed into a hybrid lot, and retested for arming and non-arming prior to being subjected to the final static non arm test. Static non arm test failures in the Hybrid lot are considered rejected and are not subject to further rework. The accuracy of the inspection and inspection equipment shall be verified hourly as a minimum, using standards containing critical defects. These standards should be readily identifiable to ensure their recovery, but should be constructed in such a manner that they are not easily recongized by congnizant inspection personnel responsible for their detection. In case the standard is missed and not classed as a reject, all material inspected and accepted since the last acceptable standard verification check shall be subjected to reinspection. The test shall be performed as specified in

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4.4.11 using equipment in accordance with 4.3.4. This equipment shall be capable of automatically segregating accepted product from rejects by sensing the presence or absence of required phosphorescent paint in the scribe lines, while not being dependent on operator observation.

Page 21 - Add paragraph: 4.3.3.13

4.3.3.13 Leakage Test, Major Defect Code No. 25009. The fuze shall be tested less pellets, cup booster and cup booster lead. The test shall be performed on twenty-five (25) bagged and sealed fuzes. Any one sealed fuze which leaks shall be cause for lot rejection. Leakage is evidenced by the collapse of the polyethylene bag after test. The test shall be performed as specified in 4.4.12 using equipment in accordance with 4.3.4. No more than five fuzes at a time shall be tested.

Page 21 - Add paragraph 4.3.3.14:

4.3.3.14 Transportation Vibration Test.-Safety failure Critical defect Code No. 25010, Operability deficiency - Major defect Code No. 25011.-One hundred fifty (150) samples or eighty (80) after 3 consecutive production lots have passed ballistic testing shall be selected for this test. The test samples will be those selected for the proving ground testing. The Transportation Vibration test shall be performed as specified in 4.4.13. Criteria for passing the test will be that prior to shipment to the Proving Ground the samples shall be x-rayed and will be in the unarmed position as evidenced by the detent pins being within a template circle whose maximum diameter does not exceed 0.859 inches. The accuracy of the inspection shall be verified by placing standards with the fuzes to be x-rayed. One standard shall contain a missing detent, and the other shall contain a detent which protrudes beyond the maximum diameter of the detent circle of 0.859 inches. Failure to properly disposition the standards on the x-ray shall result in rejection of the ballistic sample. In addition, no metal parts shall be broken, nor evidence of loose explosive present. One (1) failure shall be cause for lot rejection.

Page 21 - After 4.3.3.14 Add the following Table

TABLE II
Disposition of Tested Samples

Test No.	Test Title	Scrap	Return to Lot
4.3.3.1	Spring Embrittlement	X	
4.3.3.2	Non-arming rotor test		X
4.3.3.3	Arming test of rotor housing assembly		X
4.3.3.4	Radius at mouth of detent cavities	X	

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4.3.3.5	Load test of support cup	X	
4.3.3.6	Test of booster cup	X	
4.3.3.7	M42 Detonator test	X	
4.3.3.8	Porosity	X	
4.3.3.9	Jolt and Jumble	X	
4.3.3.10	Ballistic Testing	N/A	N/A
4.3.3.11	Firing pin assembly push test	X	
4.3.3.12	Static Non-Arm Test		X
4.3.3.13	Leakage Test		X
4.3.3.14	Transportation Vibration test		Forward to Proving Ground if acceptable

Page 22 - Para. 4.4.2

Delete sentence "...Non arming shall be evidenced by observing the red end of detonator to be aligned with the housing punch mark..."

Page 22 - Para 4.4.2 Add the following after ...the rotor housing assembly. "All rotor housing assemblies after the non-arming test shall be viewed via fluoroscope or x-ray from the top looking axially toward the base prior to assembly into the fuze body."

Page 23 - Para 4.4.9.1 Delete and substitute:

4.4.9.1 Non-Arm Check Test.-The fuze assemblies will be loaded at the Proving Ground with lead and booster explosives as per drawing 9215031 for this and subsequent ballistic tests. The ballistic sample shall then be assembled to a 57MM CTG, M306A1 (with spotting charge only), and fired from the 57MM recoilless rifle loaded to standard service velocity and pressure. The fuzes shall be fired at a one inch plywood bursting screen located 10 feet from the muzzle of the weapon.

Page 23 - Delete 4.4.9.2 and substitute

4.4.9.2 Ballistic Screen Functioning.-The ballistic sample shall then be assembled to a 57MM Ctg, M306A1 (with HE or spotting charge) and fired from 57MM recoilless rifle loaded to standard service velocity and pressure. The sample shall be fired at a one inch thick plywood bursting screen located 400 feet from the muzzle of the weapon.

Page 23 - Delete 4.4.9.3 and substitute:

4.4.9.3 Ballistic Graze Functioning.-The ballistic sample shall be assembled to a 57MM Ctg. HE, M306A1 and fired from the

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57MM recoilless rifle loaded to standard service velocity and pressure. The angle of impact shall not exceed 5 degrees on firm ground at 1000 feet.

Page 24 - Add Paragraph 4.4.11:

4.4.11 Static Non Arm Test.-Each fuze shall be tested for the non-arm condition (safe) as evidenced by the presence of phosophorescent green paint in the matched scribe lines. The fuzes shall be passed through the test equipment which will sense and segregate the safe non-armed fuzes from those armed.

Page 24 - Add Paragraph 4.4.12:

4.4.12 Leakage Test.-Each of the selected samples shall be tested for leakage by placing the sealed fuze in a vacuum chamber.The cha the sample. The absolute pressure of the air in the test chamber shall be reduced to eight (8) inches of mercury, maximum.

Page 24 - Add paragraph 4.4.13:

4.4.13 Transportation Vibration Test.-The sample designated for ballistic test shall be assembled with lead cup covered by the booster disc, and with a booster cup to maintain the fuze seal. The sample shall then be subjected to transportation vibration in accordance with MIL-STD-331, Test 104, at the contractors plant. After TV testing, the sample shall undergo X-ray inspection per requirements of para. 4.3.3.14 to assure the fuze is safe for shipment to the proving ground. Units shall be forwarded to the proving ground upon approval of the procuring activity.

Page 24 - 5.1 Delete "Level A"
Substitute "Level C"

Page 24 - Add paragraph 5.1.1:

5.1.1 Booster Component Parts.-Booster cup, booster lead, and closing disc shall be packaged in accordance with MIL-STD-1169 instructions.

Page 24 - Add paragraph 5.1.2:

5.1.2 Fuze Assembly.-The fuze assembly shall be packaged in accordance with drawing 9215630.

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Page 24 Paragraph 5.2 Delete "Level A"
Substitute "Level C"

Page 24 - Paragraph 5.3 To end of Paragraph 5.3 Marking, add "as applicable".

Custodian
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Preparing Activity
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Projectile No. 1390-A-001

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